

# Topic 08: Bernoulli Equations & Composition Substitutions

## Professor Leonard Master Series · Lessons 22 to 24.5

**Course:** Applied Ordinary Differential Equations (ENGR 213 Companion)

### 1. Whiteboard Intuition: Taming Non-Linear Powers

Professor Leonard introduces Bernoulli equations with high drama:

*"Look at this equation:  $y' + P(x)y = Q(x)y^n$ . It is so close to being a friendly linear differential equation, but that nasty  $y^n$  on the right ruins everything! James Bernoulli showed us that if we divide by  $y^n$  and substitute  $u = y^{1-n}$ , the non-linear power collapses, the derivative aligns perfectly, and the equation magically turns into a 100% standard linear ODE in  $u$ !"*

$$y' + P(x)y = Q(x)y^n$$

Divide by  $y^n$ :

$$y^{-n}y' + P(x)y^{1-n} = Q(x)$$

$$\text{Let } u = y^{1-n} \implies \frac{du}{dx} = (1-n)y^{-n}\frac{dy}{dx} \implies y^{-n}y' = \frac{1}{1-n}u'$$

$$\frac{1}{1-n}u' + P(x)u = Q(x) \implies u' + (1-n)P(x)u = (1-n)Q(x)$$

### 2. Professor Leonard's Whiteboard Problem Walkthroughs

#### Problem 8.1: Classic Bernoulli ( $n = 2$ )

**Statement:** Solve  $\frac{dy}{dx} - y = e^x y^2$ .

**Step-by-Step Whiteboard Solution:**

- **Step 1: Identify Bernoulli Form:**  $P(x) = -1, Q(x) = e^x, n = 2$ .
- **Step 2: Divide by  $y^2$ :**

$$y^{-2}\frac{dy}{dx} - y^{-1} = e^x$$

- **Step 3: Define  $u$  and  $du/dx$ :**

$$u = y^{1-2} = y^{-1} \implies \frac{du}{dx} = -y^{-2}\frac{dy}{dx} \implies y^{-2}\frac{dy}{dx} = -\frac{du}{dx}$$

- **Step 4: Substitute into ODE:**

$$-\frac{du}{dx} - u = e^x \implies \frac{du}{dx} + u = -e^x$$

- **Step 5: Solve Linear ODE for  $u(x)$ :**

$$\mu(x) = e^{\int 1 dx} = e^x$$

$$\frac{d}{dx}[e^x u] = e^x(-e^x) = -e^{2x}$$

$$e^x u = -\int e^{2x} dx = -\frac{1}{2}e^{2x} + C$$

$$u(x) = -\frac{1}{2}e^x + Ce^{-x}$$

- **Step 6: Back-substitute  $u = 1/y$ :**

$$\frac{1}{y} = -\frac{1}{2}e^x + Ce^{-x} = \frac{2Ce^{-x} - e^x}{2} = \frac{C_1 e^{-x} - e^x}{2}$$

$$y(x) = \frac{2}{C_1 e^{-x} - e^x}$$

(Also note singular equilibrium solution  $y \equiv 0$ )

## Problem 8.2: Bernoulli ( $n = 3$ ) with Polynomial Coefficients

**Statement:** Solve  $x \frac{dy}{dx} + y = xy^3$ .

**Step-by-Step Whiteboard Solution:**

- **Step 1: Standardize:**

$$\frac{dy}{dx} + \frac{1}{x}y = y^3 \quad (n = 3)$$

- **Step 2: Divide by  $y^3$ :**

$$y^{-3} \frac{dy}{dx} + \frac{1}{x}y^{-2} = 1$$

- **Step 3: Substitute  $u = y^{1-3} = y^{-2}$ :**

$$\frac{du}{dx} = -2y^{-3} \frac{dy}{dx} \implies y^{-3} \frac{dy}{dx} = -\frac{1}{2} \frac{du}{dx}$$

- **Step 4: Substitute:**

$$-\frac{1}{2} \frac{du}{dx} + \frac{1}{x} u = 1 \implies \frac{du}{dx} - \frac{2}{x} u = -2$$

- **Step 5: Integrating Factor:**

$$\mu(x) = e^{\int -\frac{2}{x} dx} = e^{-2 \ln x} = x^{-2} = \frac{1}{x^2}$$

$$\frac{d}{dx} \left[ \frac{u}{x^2} \right] = -\frac{2}{x^2} = -2x^{-2}$$

$$\frac{u}{x^2} = \int -2x^{-2} dx = 2x^{-1} + C = \frac{2}{x} + C$$

$$u(x) = 2x + Cx^2$$

- **Step 6: Back-substitute**  $u = y^{-2} = 1/y^2$ :

$$\frac{1}{y^2} = 2x + Cx^2 \implies y^2 = \frac{1}{2x + Cx^2} \implies y(x) = \pm \frac{1}{\sqrt{2x + Cx^2}}$$

### Problem 8.3: Linear Combination Substitution $y' = f(Ax + By + C)$

**Statement:** Solve  $\frac{dy}{dx} = (x + y + 3)^2$ .

**Step-by-Step Whiteboard Solution:**

- **Step 1: Recognize Linear Argument:** The argument inside the square is  $x + y + 3$ .
- **Step 2: Substitute**  $u = x + y + 3$ :

$$\frac{du}{dx} = 1 + \frac{dy}{dx} \implies \frac{dy}{dx} = \frac{du}{dx} - 1$$

- **Step 3: Substitute into ODE:**

$$\frac{du}{dx} - 1 = u^2 \implies \frac{du}{dx} = u^2 + 1$$

- **Step 4: Separate variables:**

$$\frac{du}{u^2 + 1} = dx$$

- **Step 5: Integrate:**

$$\arctan(u) = x + C \implies u(x) = \tan(x + C)$$

- **Step 6: Back-substitute**  $u = x + y + 3$ :

$$x + y + 3 = \tan(x + C) \implies y(x) = \tan(x + C) - x - 3$$

### Problem 8.4: Trigonometric Linear Argument

**Statement:** Solve  $\frac{dy}{dx} = \sin^2(x - y + 1)$ .

**Step-by-Step Whiteboard Solution:**

- **Step 1: Let  $u = x - y + 1$ :**

$$\frac{du}{dx} = 1 - \frac{dy}{dx} \implies \frac{dy}{dx} = 1 - \frac{du}{dx}$$

- **Step 2: Substitute:**

$$1 - \frac{du}{dx} = \sin^2 u \implies \frac{du}{dx} = 1 - \sin^2 u = \cos^2 u$$

- **Step 3: Separate:**

$$\frac{du}{\cos^2 u} = dx \implies \sec^2 u du = dx$$

- **Step 4: Integrate:**

$$\tan(u) = x + C$$

- **Step 5: Back-substitute:**

$$\tan(x - y + 1) = x + C \implies x - y + 1 = \arctan(x + C) \implies y(x) = x + 1 - \arctan(x + C)$$

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### 3. Common Exam Traps & Professor Leonard Warnings

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- **Trap 1: Dropping the  $(1 - n)$  Factor:** Forgetting that  $du/dx = (1 - n)y^{-n}y'$ . If  $n = 3$ , that factor of  $-2$  must multiply the RHS!
- **Trap 2: Forgetting Singular Solution  $y = 0$ :** When dividing by  $y^n$ , you assumed  $y \neq 0$ . Check if  $y \equiv 0$  solves the original ODE.